

Exp No:16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

PROGRAM:

```
#include <reg51.h>
```

```
#include <stdio.h>
```

```
char rx_char;
```

```
bit new_data = 0;
```

```
char putchar(char c)
```

```
{
```

```
    SBUF = c;
```

```
    while(TI == 0);
```

```
    TI = 0;
```

```
    return c;
```

```
}
```

```
void serial_ISR(void) interrupt 4
```

```
{
```

```
    if(RI)
```

```
    {
```

```
        RI = 0;
```

```
        rx_char = SBUF;
```

```
        new_data = 1;
```

```
    }
```

```
}
```

```
void delay(void)
```

```
{m
```

```

    unsigned int i,j;
    for(i=0;i<100;i++)
        for(j=0;j<500;j++);
}

void main(void)
{
    SCON = 0x50;
    TMOD = 0x20;
    TH1 = 0xFD;
    TL1 = 0xFD;
    TR1 = 1;

    IE = 0x90;

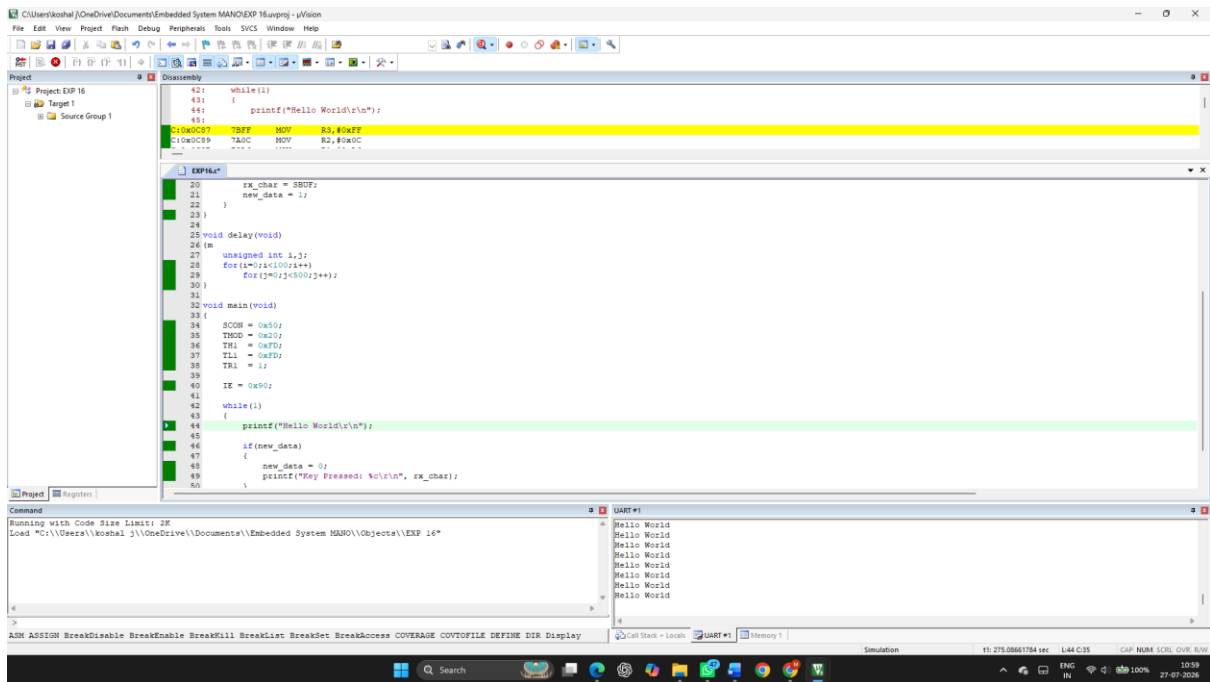
    while(1)
    {
        printf("Hello World\r\n");

        if(new_data)
        {
            new_data = 0;
            printf("Key Pressed: %c\r\n", rx_char);
        }

        delay();
    }
}

```

1) Displaying a Message:



2)During interruption:

